

# DB Normalization

Sistemas de Informação 1

# Credits and Disclaimer

- This material/slides are adapted from:
  - SINF1 2024/25 slides produced by Telmo Matos
  - Books:
    - Database Systems - A Practical Approach to Design, Implementation, and Management, Thomas Connolly & Carolyn Begg; Pearson 2015; 6th Edition (Global Ed.); ISBN: 978-1-292-06118-4

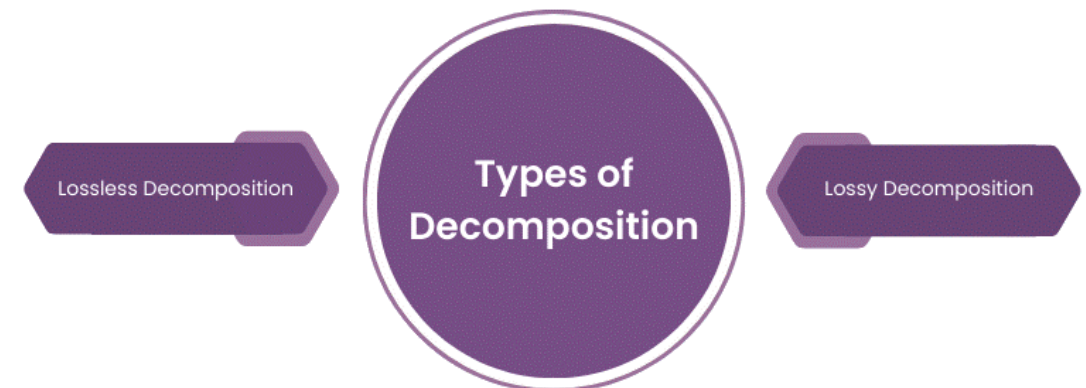


# DB Normalization

- A systematic process that decomposes a database model into smaller tables with stronger relationships between them, to **reduce data redundancy** and **improve data integrity**.
- All attributes of the original schema **must be preserved** in the resulting schemas.

# DB Normalization

- The decomposition of a table must satisfy the following properties:
  - **Lossless decomposition** – The resulting schema must contain the same information as the original schema.
  - **Dependency preservation** – The resulting tables must allow the enforcement of all functional dependencies present in the original table.

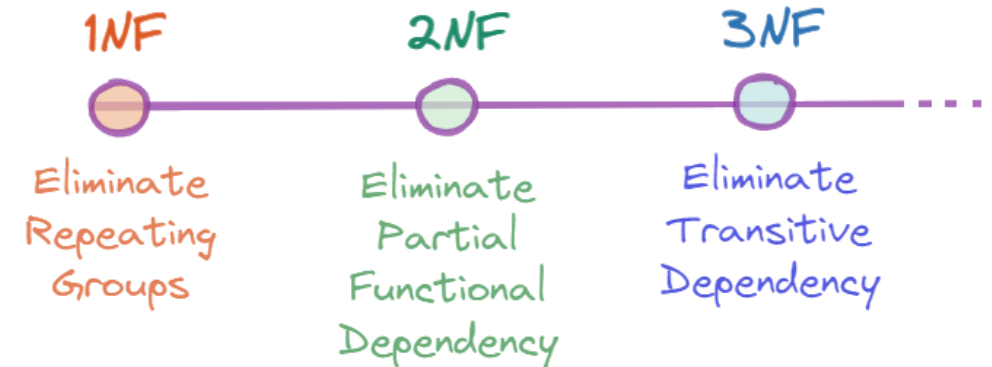


# DB Normalization

- In a normalization process:
  - Each table represents a **single subject**.
  - **No data item** is stored unnecessarily in more than one table.
  - All attributes in a table **depend on the primary key**.
  - Each table is free from **insertion, update and deletion anomalies**.
- Tables are progressively decomposed based on the identified dependencies.

# DB Normalization

- Database normalization is performed through a series of steps called **Normal Forms**, which are applied successively:
  - First Normal Form (1NF)
  - Second Normal Form (2NF)
  - Third Normal Form (3NF)
  - ...



# DB Normalization Anomalies

- Database normalization anomalies are problems that arise from poorly designed databases, leading to data redundancy and inconsistencies. These issues can be minimized through normalization.
- Types of Anomalies:
  - **Insertion Anomaly** - Difficulty in adding new data without introducing irrelevant or redundant information.
  - **Deletion Anomaly** - Loss of important data when deleting a record, which may also remove other related information.
  - **Update Anomaly** - Inconsistencies that occur when updating data stored in multiple locations.

## DB Normalization Anomalies

- **Insertion:** If you want to add a new course (e.g., “*LAB*”) that currently has no enrolled students, you will need to insert a row with dummy student data (*student name*, *year*, and *semester*).

| Student           | Course               | Year | Semester |
|-------------------|----------------------|------|----------|
| António           | SINF1                | 2    | 2        |
| Raquel            | SINF2                | 3    | 1        |
| Vitor             | Operational Research | 2    | 2        |
| José              | SINF1                | 2    | 2        |
| <i>Dummy name</i> | LAB                  | 3    | 2        |



# DB Normalization Anomalies

- **Deletion:** By removing student Raquel, I am also removing the SINF2 course.

| Student | Course               | Year         | Semester     |
|---------|----------------------|--------------|--------------|
| António | SINF1                | 2            | 2            |
| Raquel  | <del>SINF2</del>     | <del>3</del> | <del>1</del> |
| Vitor   | Operational Research | 2            | 2            |
| José    | SINF1                | 2            | 2            |

## DB Normalization Anomalies

- **Update:** Changing the course name from *SINF1* to *Information Systems 1* requires updating multiple entries; otherwise, it may cause inconsistencies.

| Student | Course                | Year | Semester |
|---------|-----------------------|------|----------|
| António | Information Systems 1 | 2    | 2        |
| Raquel  | SINF2                 | 3    | 1        |
| Vitor   | Operational Research  | 2    | 2        |
| José    | Information Systems 1 | 2    | 2        |

## Example: A Database Requiring Normalization

- In a company that manages construction projects:
  - Clients are charged for the hours worked on each contract;
  - The hourly rate depends on each employee's category;
  - A report is produced periodically containing the information shown in the following table:



# Example: A Database Requiring Normalization

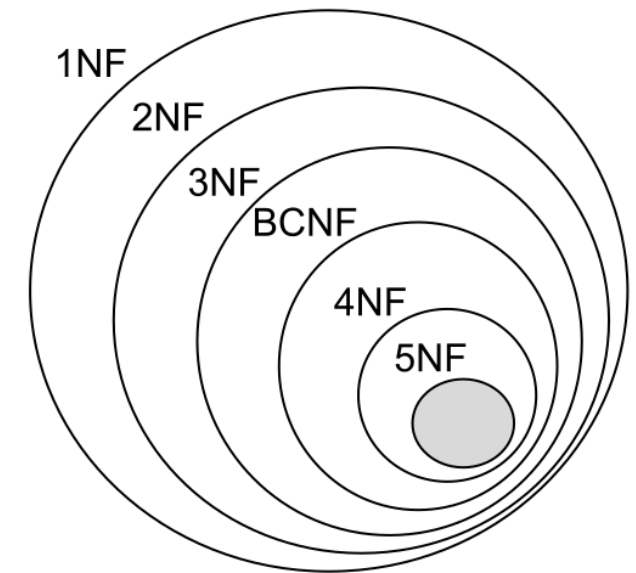
| PROJECT NUMBER | PROJECT NAME | EMPLOYEE NUMBER | EMPLOYEE NAME        | JOB CLASS             | CHARGE/HOUR | HOURS BILLED | TOTAL CHARGE       |
|----------------|--------------|-----------------|----------------------|-----------------------|-------------|--------------|--------------------|
| 15             | Evergreen    | 103             | June E. Arbough      | Elec. Engineer        | \$ 85.50    | 23.8         | \$ 2,034.90        |
|                |              | 101             | John G. News         | Database Designer     | \$105.00    | 19.4         | \$ 2,037.00        |
|                |              | 105             | Alice K. Johnson *   | Database Designer     | \$105.00    | 35.7         | \$ 3,748.50        |
|                |              | 106             | William Smithfield   | Programmer            | \$ 35.75    | 12.6         | \$ 450.45          |
|                |              | 102             | David H. Senior      | Systems Analyst       | \$ 96.75    | 23.8         | \$ 2,302.65        |
|                |              |                 |                      | <b>Subtotal</b>       |             |              | <b>\$10,573.50</b> |
| 18             | Amber Wave   | 114             | Annelise Jones       | Applications Designer | \$ 48.10    | 25.6         | \$ 1,183.26        |
|                |              | 118             | James J. Frommer     | General Support       | \$ 18.36    | 45.3         | \$ 831.71          |
|                |              | 104             | Anne K. Ramoras *    | Systems Analyst       | \$ 96.75    | 32.4         | \$ 3,134.70        |
|                |              | 112             | Darlene M. Smithson  | DSS Analyst           | \$ 45.95    | 45.0         | \$ 2,067.75        |
|                |              |                 |                      | <b>Subtotal</b>       |             |              | <b>\$ 7,265.52</b> |
| 22             | Rolling Tide | 105             | Alice K. Johnson     | Database Designer     | \$105.00    | 65.7         | \$ 6,998.50        |
|                |              | 104             | Anne K. Ramoras      | Systems Analyst       | \$ 96.75    | 48.4         | \$ 4,682.70        |
|                |              | 113             | Delbert K. Joenbrood | Applications Designer | \$ 48.10    | 23.6         | \$ 1,135.16        |
|                |              | 111             | Geoff B. Wabash      | Clerical Support      | \$ 26.87    | 22.0         | \$ 591.14          |
|                |              | 106             | William Smithfield   | Programmer            | \$ 35.75    | 12.8         | \$ 457.60          |
|                |              |                 |                      | <b>Subtotal</b>       |             |              | <b>\$13,765.10</b> |
| 25             | Starflight   | 107             | Maria D. Alonzo      | Programmer            | \$ 35.75    | 25.6         | \$ 915.20          |
|                |              | 115             | Travis B. Bawangi    | Systems Analyst       | \$ 96.75    | 45.8         | \$ 4,431.15        |
|                |              | 101             | John G. News *       | Database Designer     | \$105.00    | 56.3         | \$ 5,911.50        |
|                |              | 114             | Annelise Jones       | Applications Designer | \$ 48.10    | 33.1         | \$ 1,592.11        |
|                |              | 108             | Ralph B. Washington  | Systems Analyst       | \$ 96.75    | 23.6         | \$ 2,283.30        |
|                |              | 118             | James J. Frommer     | General Support       | \$ 18.36    | 30.5         | \$ 559.98          |
|                |              | 112             | Darlene M. Smithson  | DSS Analyst           | \$ 45.95    | 41.4         | \$ 1,902.33        |
|                |              |                 |                      | <b>Subtotal</b>       |             |              | <b>\$17,595.57</b> |
|                |              |                 |                      | <b>Total</b>          |             |              | <b>\$49,199.69</b> |

Note: \* indicates project leader

# DB Normalization

## ■ Normal Forms:

- Proposed by Codd in 1970 to ensure the absence of anomalies in the tables;
- Rules that define different stages of database normalization, aiming to minimize redundancy and improve data integrity.



# First Normal Form (1NF)

- In 1NF, every database cell contains an atomic value that **cannot be further divided**; in other words, the relation should not have multivalued attributes.
- Conversion to 1NF:
  - Relational tables **must not contain repeating groups**;
  - Relational tables **must contain only atomic values**;
  - Normalizing the table structure reduces data redundancy.

# Conversion to 1NF

| Course    | Code | ECTS | Program  |
|-----------|------|------|----------|
| Databases | DB   | 5    | 231, 240 |
| Coding    | CP   | 4    | 231      |



| Course    | Sigla | ECTS | Program |
|-----------|-------|------|---------|
| Databases | DB    | 5    | 231     |
| Databases | DB    | 5    | 240     |
| Coding    | CP    | 4    | 231     |

| ID        | Name         | Address                          |
|-----------|--------------|----------------------------------|
| 123456789 | José Cunha   | Rua das Oliveiras, nº4, 4715-343 |
| 222333444 | Carlos Costa | Avenida Central, nº 80, 4700-234 |



| ID        | Name         | Street            | Door | Postal Code |
|-----------|--------------|-------------------|------|-------------|
| 123456789 | José Cunha   | Rua das Oliveiras | 4    | 4715-343    |
| 222333444 | Carlos Costa | Avenida Central   | 80   | 4700-234    |

## Conversion to 1NF

### ■ Delete repeating groups:

- Display data in a tabular format where each cell contains a single value and there are no repeating groups.
- Eliminate repeating groups by removing null values and ensuring that each attribute in a repeating group contains an appropriate data value



# DB Normalization

| PROJECT NUMBER | PROJECT NAME | EMPLOYEE NUMBER | EMPLOYEE NAME        | JOB CLASS             | CHARGE/HOUR | HOURS BILLED | TOTAL CHARGE       |
|----------------|--------------|-----------------|----------------------|-----------------------|-------------|--------------|--------------------|
| 15             | Evergreen    | 103             | June E. Arbough      | Elec. Engineer        | \$ 85.50    | 23.8         | \$ 2,034.90        |
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|                |              | 114             | Annelise Jones       | Applications Designer | \$ 48.10    | 33.1         | \$ 1,592.11        |
|                |              | 108             | Ralph B. Washington  | Systems Analyst       | \$ 96.75    | 23.6         | \$ 2,283.30        |
|                |              | 118             | James J. Frommer     | General Support       | \$ 18.36    | 30.5         | \$ 559.98          |
|                |              | 112             | Darlene M. Smithson  | DSS Analyst           | \$ 45.95    | 41.4         | \$ 1,902.33        |
|                |              |                 |                      | <b>Subtotal</b>       |             |              | <b>\$17,595.57</b> |
|                |              |                 |                      | <b>Total</b>          |             |              | <b>\$49,199.69</b> |

Note: \* indicates project leader

# DB Normalization – 1NF

| PROJ_NUM | PROJ_NAME    | EMP_NUM | EMP_NAME               | JOB_CLASS             | CHG_HOUR | HOURS |
|----------|--------------|---------|------------------------|-----------------------|----------|-------|
| 15       | Evergreen    | 103     | June E. Arbough        | Elect. Engineer       | 84.50    | 23.8  |
| 15       | Evergreen    | 101     | John G. News           | Database Designer     | 105.00   | 19.4  |
| 15       | Evergreen    | 105     | Alice K. Johnson *     | Database Designer     | 105.00   | 35.7  |
| 15       | Evergreen    | 106     | William Smithfield     | Programmer            | 35.75    | 12.6  |
| 15       | Evergreen    | 102     | David H. Senior        | Systems Analyst       | 96.75    | 23.8  |
| 18       | Amber Wave   | 114     | Annelise Jones         | Applications Designer | 48.10    | 24.6  |
| 18       | Amber Wave   | 118     | James J. Frommer       | General Support       | 18.36    | 45.3  |
| 18       | Amber Wave   | 104     | Anne K. Ramoras *      | Systems Analyst       | 96.75    | 32.4  |
| 18       | Amber Wave   | 112     | Darlene M. Smithson    | DSS Analyst           | 45.95    | 44.0  |
| 22       | Rolling Tide | 105     | Alice K. Johnson       | Database Designer     | 105.00   | 64.7  |
| 22       | Rolling Tide | 104     | Anne K. Ramoras        | Systems Analyst       | 96.75    | 48.4  |
| 22       | Rolling Tide | 113     | Delbert K. Joenbrood * | Applications Designer | 48.10    | 23.6  |
| 22       | Rolling Tide | 111     | Geoff B. Wabash        | Clerical Support      | 26.87    | 22.0  |
| 22       | Rolling Tide | 106     | William Smithfield     | Programmer            | 35.75    | 12.8  |
| 25       | Starflight   | 107     | Maria D. Alonzo        | Programmer            | 35.75    | 24.6  |
| 25       | Starflight   | 115     | Travis B. Bawangi      | Systems Analyst       | 96.75    | 45.8  |
| 25       | Starflight   | 101     | John G. News *         | Database Designer     | 105.00   | 56.3  |
| 25       | Starflight   | 114     | Annelise Jones         | Applications Designer | 48.10    | 33.1  |
| 25       | Starflight   | 108     | Ralph B. Washington    | Systems Analyst       | 96.75    | 23.6  |
| 25       | Starflight   | 118     | James J. Frommer       | General Support       | 18.36    | 30.5  |
| 25       | Starflight   | 112     | Darlene M. Smithson    | DSS Analyst           | 45.95    | 41.4  |

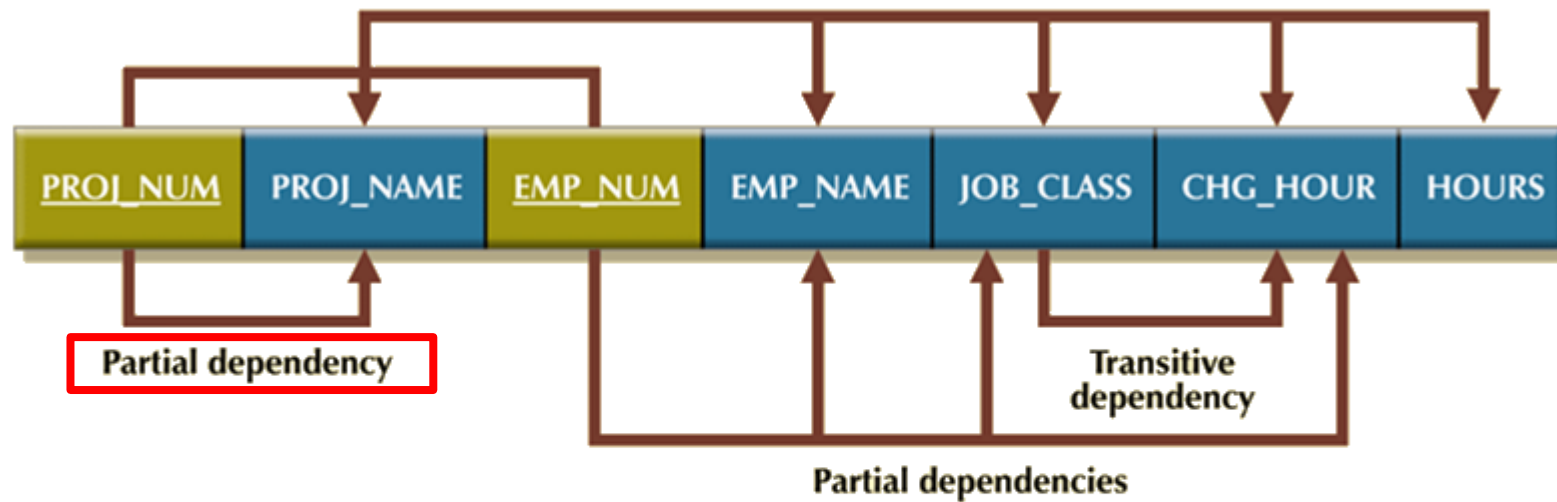
## Second Normal Form (2NF)

- In 2NF, a relation must first be in 1NF, and no partial dependency should exist.
  - A partial dependency occurs when a non-prime attribute depends on only part of a composite primary key.

## Conversion to 2NF

- Identify the primary key.
  - The primary key must uniquely identify any value of an attribute.
- Identify all functional dependencies using a dependency diagram:
  - Shows all dependencies in each table structure;
  - Provides an overview of the relationships between attributes
  - Helps ensure that no relevant dependency is overlooked

# Dependency diagram



1NF (PROJ\_NUM, EMP\_NUM, PROJ\_NAME, EMP\_NAME, JOB\_CLASS, CHG\_HOURS, HOURS)

PARTIAL DEPENDENCIES:

(PROJ\_NUM  $\Rightarrow$  PROJ\_NAME)

(EMP\_NUM  $\Rightarrow$  EMP\_NAME, JOB\_CLASS, CHG\_HOUR)

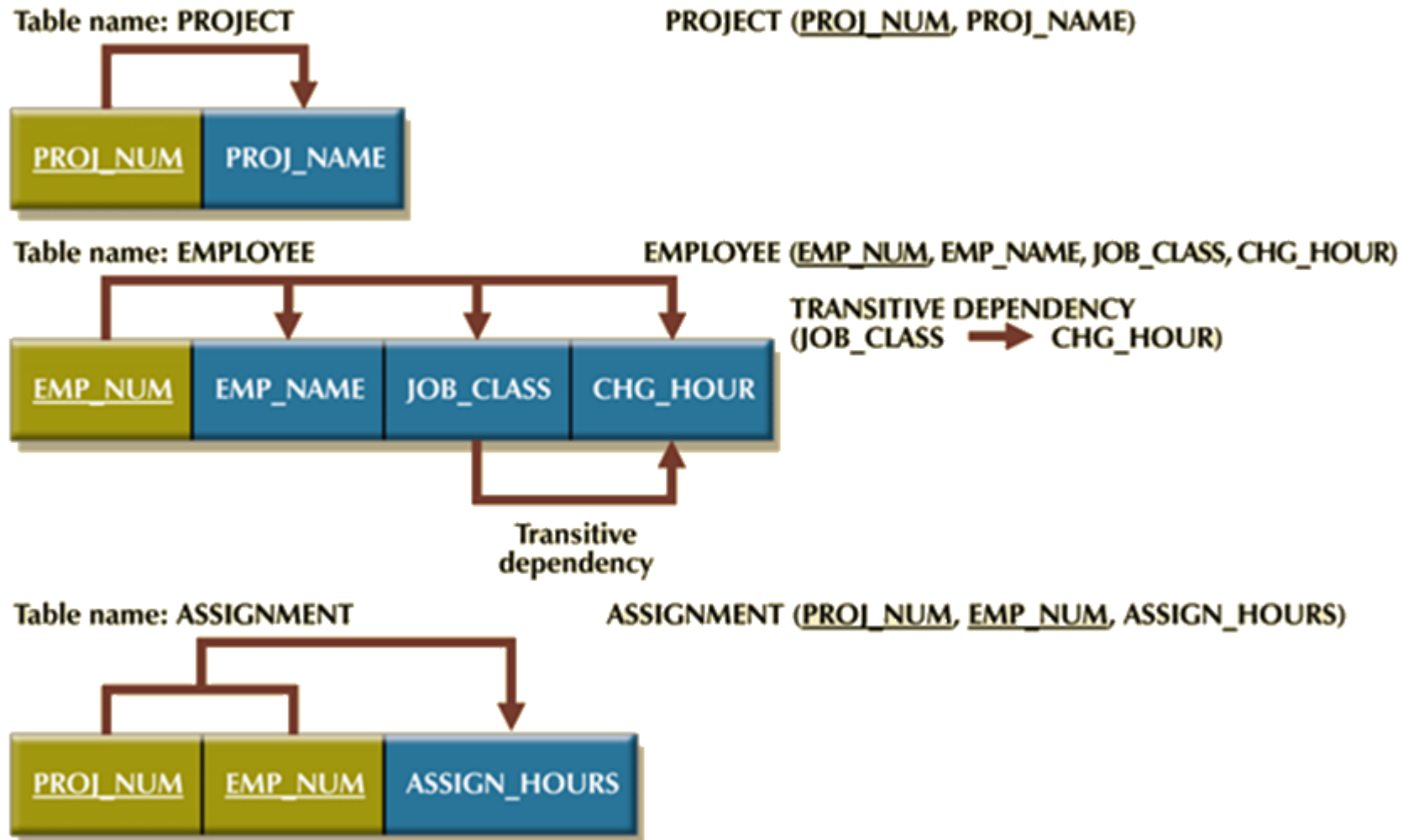
TRANSITIVE DEPENDENCY:

(JOB CLASS  $\Rightarrow$  CHG\_HOUR)

# Conversion to 2NF

- **Step 1: Identify all key components**
  - Write each component of the key on a separate line, then write the original (composite) key on the last line;
  - Each identified component will be key in a new table.
- **Step 2: Identify dependent attributes**
  - Determine which attributes depend on the elements identified in Step 1.
  - At this stage, most anomalies are eliminated...

# Conversion to 2NF



## Third Normal Form (3NF)

- In 3NF, a relation **must first be in 2NF**, and **no transitive dependency** should exist; that is, non-prime attributes should not determine other non-prime attributes.



## Conversion to 3NF

- Step 1: Identify each new determinant
  - For each transitive dependency, write its determinant as the primary key of a new table.
    - Determinant: Any attribute whose value determines other values in the same row.
- Step 2: Identify dependent attributes
- Step 3: Remove dependent attributes from transitive dependencies

## Conversion to 3NF

- Step 1: Identify each new determinant
- Step 2: Identify dependent attributes
  - Identify which attributes depend on each key component and note the dependencies;
  - Name each table appropriately according to its content and function.
- Step 3: Remove dependent attributes from transitive dependencies

## Conversion to 3NF

- Step 1: Identify each new determinant
- Step 2: Identify dependent attributes
- **Step 3: Remove dependent attributes from transitive dependencies**
  - Remove all attributes involved in transitive dependencies;
  - Create a new dependency diagram showing all tables defined in Steps 1–3;
  - Verify the new and modified tables to ensure that each has a well-defined key and contains no inappropriate dependencies.

# Conversion to 3NF



Table name: PROJECT

PROJECT (PROJ\_NUM, PROJ\_NAME)



Table name: JOB

JOB (JOB\_CLASS, CHG\_HOUR)

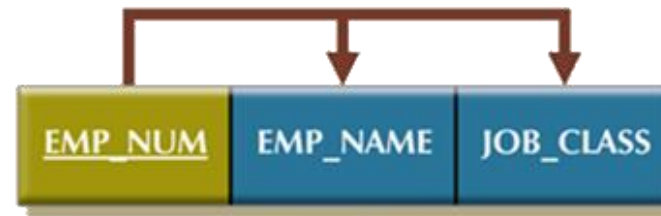


Table name: EMPLOYEE

EMPLOYEE (EMP\_NUM, EMP\_NAME, JOB\_CLASS)



Table name: ASSIGNMENT

ASSIGNMENT (PROJ\_NUM, EMP\_NUM, ASSIGN\_HOURS)

## Other Normal Forms

### ■ Boyce-Codd Normal Form (BCNF)

- A relation is in BCNF if for every functional dependency  $X \rightarrow Y$ ,  $X$  is a superkey.
  - A superkey is a set of one or more attributes in a relation (table) that uniquely identifies each tuple (row) in that table

### ■ Fourth Normal Form (4NF)

- A table is in 4NF if it is in BCNF and has no multi-valued dependencies.

### ■ Fifth Normal Form (5NF)

- A table is in 5NF if it is in 4NF and cannot be further decomposed without losing information (i.e., all decompositions are lossless and necessary).

### ■ Sixth Normal Form (6NF)

- A relation is in 6NF if it is in 5NF and every join dependency is trivial.